

GEOMETRY-INFORMED SYNTHETIC-TO-REAL LOCALIZATION

Teach with geometry. Inspect with a photograph.

Can known 3D damage in simulated blades help a model find damage in real photographs—with fewer real annotations?

SUPERVISOR & TEAM DEMONSTRATION

A REAL SAVED OUTPUT · AEBIS 329



● Saved prediction ● Reference boundary

56.43% mask overlap on this image. Existing real-only model; no demonstrated geometry benefit.

We have built

Data pipelines, synthetic labels, trained models and checked examples.

We still need to establish

Whether geometry improves real defect masks under a fair comparison.

This proposal tests

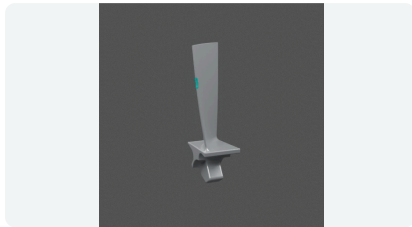
The value of extra 3D teaching information at fixed real-label budgets.

SAY IT SIMPLY “We want to teach using information available in simulation, then inspect a real blade from a normal photograph.”

From known synthetic damage to a real defect mask

The 3D information helps teach the model. The deployed model receives an RGB photograph.

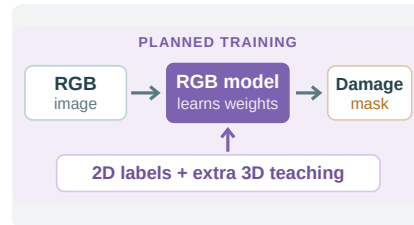
1



Create teaching data

Rendered images, damage masks and verified links to the simulated surface.

2



Train and adapt

Compare matched training conditions, then use exactly the same real masks.

3



Inspect a photograph

Use only the real image. No CAD, depth map or healthy reference is required.

4



Return a damage mask

Highlight suspected damaged pixels and test them against human labels.

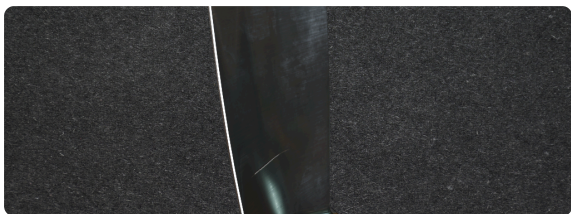
1 · We know where the simulated damage is

A verified synthetic sample can supply both damaged pixels and their 3D surface locations. The broad static collection supplies ordinary image masks.

Images above are existing saved examples showing the workflow's input and output format. The final image is an existing real-only prediction, not a result from the proposed geometry-trained model.

Three useful data pools. Different kinds of supervision.

An image, a damage mask and a verified 3D link are different assets.



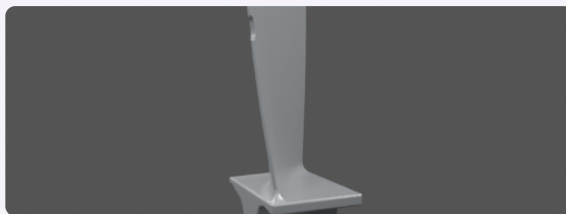
BROAD STATIC SYNTHETIC COLLECTION

12,344

Accepted RGB images

2,500 healthy + 9,844 damaged.
Appearance and 2D mask supervision.

The static archive does not have verified camera/CAD links for every image.



GEOMETRY-LINKED CANDIDATE POOL

582

Accepted records in the proposal

571 pixel-distinct RGBs.
Includes nick, dent and control cases.

Eligibility needs curation. Records and views are not independent physical blades.



LABELLED REAL AEBIS PHOTOGRAPHS

397 + 131

Training + reused development

528 damaged images with labels.
No healthy images in this pool.

59 reserved archive images need a freshness audit. A new evaluation set remains necessary.

Other real images AeBAD: 2,160 stills + 3,410 video frames; previous test exposure must remain disclosed.

Personal photographs 174 originals; 159 used. Existing blade outlines do not provide damage-mask truth.

First geometry comparison: nicks and dents. Scratches and corrosion can supply ordinary 2D supervision when label meanings agree; their geometric benefit remains untested.

We have working parts—and evidence of the difficult part.

Real photographs remain the test of whether the extra teaching information is useful.

14 SEP · BROAD SYNTHETIC

Rendering is useful, but transfer is hard

Mean IoU: **74.4%** on synthetic evaluation; **3.979%** on real AeBAD-S. Different populations.

Synthetic example · Dent_1000



Input → red-orange prediction → cyan reference. This image: **86.63% IoU**.

15 SEP · REAL LABELS

A strong real baseline emerged

Retained model: **61.941% ± 3.081** IoU on reused AEBIS development images.

17 SEP · GEOMETRY

Better point, similar mask

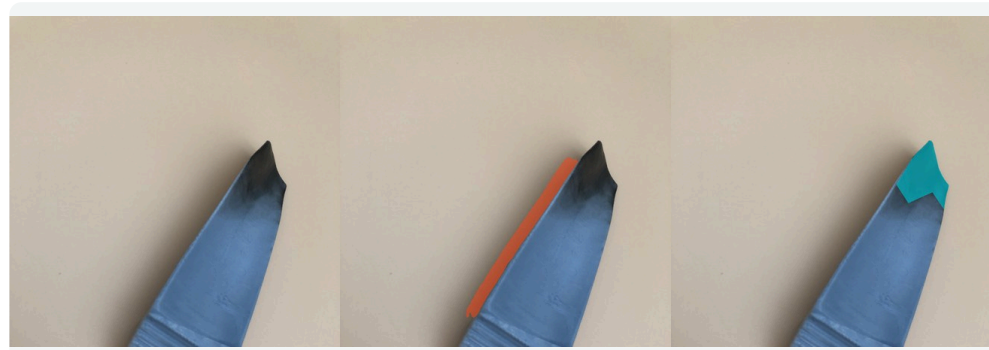
Adding 3D teaching improved synthetic point placement. Mean mask IoU fell slightly.

18–22 SEP · SUPPORTING TOOLS

Mapping and physical fitting were explored

Useful controlled examples and failure checks. Real physical recovery remains unverified.

Real example · same broad synthetic-only model



Input → red-orange prediction → cyan reference. Saved failure has **no overlap** with the damage.

Research question sharpened: will geometry improve real segmentation beyond using the same synthetic images with ordinary 2D teaching?

Historical numbers use different datasets, model stages and protocols. They explain the research direction; they are not a matched leaderboard. ± denotes variation across three model seeds.

The model finds damage—but misses part of its extent.

This is the existing real-only model, using a single photograph.

(a) Real RGB input



(b) Saved model prediction



(c) Supplied damage reference



● Amber: predicted damaged pixels ● Teal: reference damaged pixels

MASK OVERLAP · IOU

56.43%

Correct overlap ÷ union

PRECISION

97.16%

Most marked pixels are correct.

RECALL

57.37%

Much of the reference is missed.

PIXEL COUNTS

685 correct · **20** extra

509 missed

A numerical example: $\text{IoU} = 685 \div (685 + 20 + 509) = 56.43\%$. This measures overlap of damaged pixels, not the percentage of correctly classified blades.

A missed defect is as important as a good-looking overlay.

The same retained model marks no pixels in this damaged image.

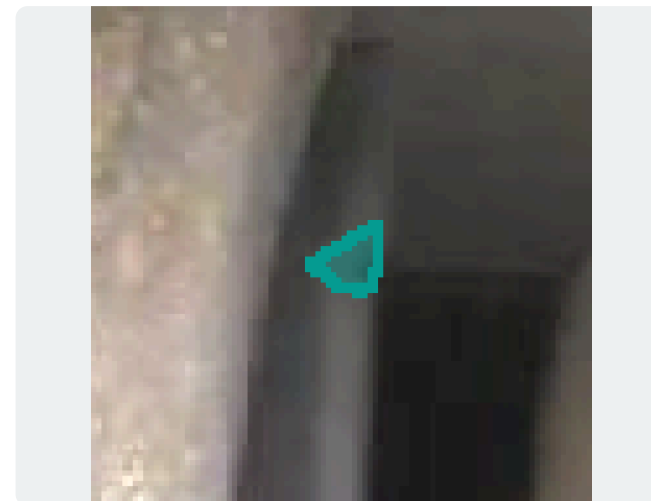
(a) Real RGB input



(b) Saved prediction · empty



(c) Reference · enlarged detail



Display-only crop chosen from the reference; model inference used the full saved grid.

REFERENCE DAMAGE

94 pixels

Damage exists in the supplied mask.

PREDICTED DAMAGE

0 pixels

The saved model misses it completely.

IOU / PRECISION / RECALL

0 / 0 / 0

Reported using the saved scoring convention.

What the proposed study must improve: finding the right pixels on real images—including small or difficult defects—without creating extra alarms on healthy blades.

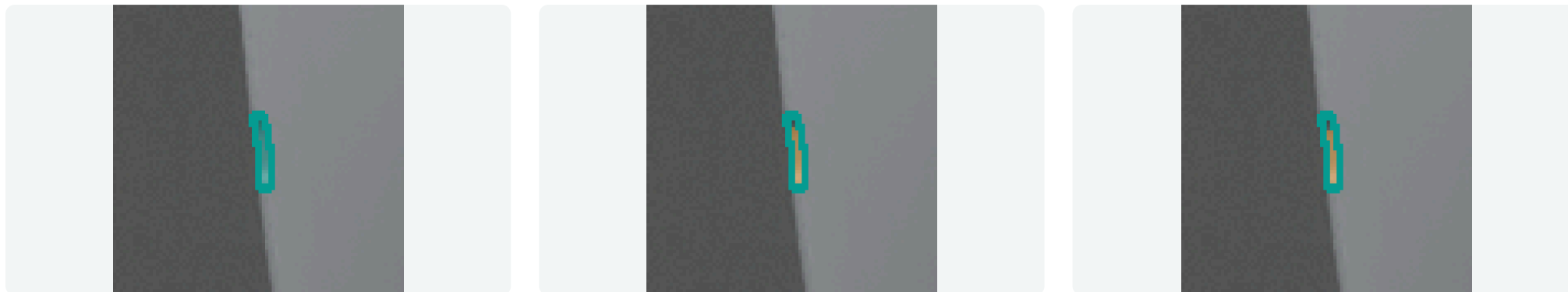
A correct 3D point does not guarantee a good damage mask.

Same RGB input, matched models, additional 3D supervision in one condition.

Reference · exposed cut surface B

2D-only teaching · 75.76% image IoU

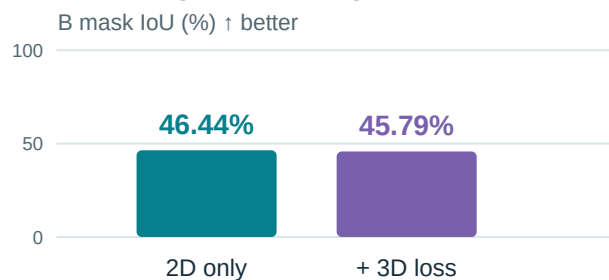
Additional 3D teaching · 68.04% image IoU



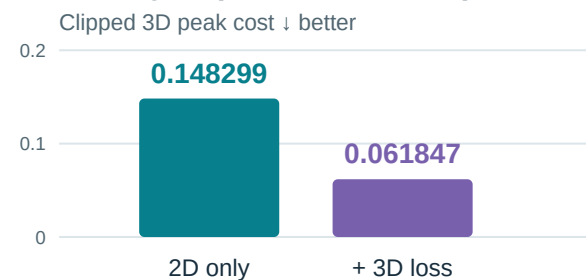
p5_s1p2_v000 · amber prediction / teal reference outline · identical reference-derived display crop · both saved peaks reach the target (zero target distance).

ACTUAL SAVED STUDY

Mask overlap did not improve



The strongest point was better placed



Three paired seeds · one CAD · 56 evaluation views · 31 B-target-eligible views

Point cost: unitless, capped at 1. These are synthetic results; real-photo benefit remains unmeasured.

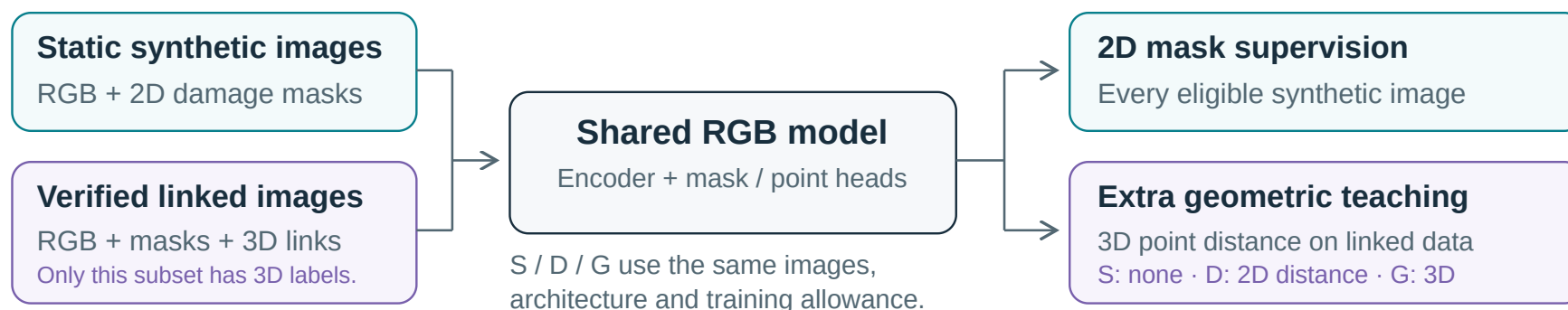
This supports better point localization in a controlled synthetic setting. A real-image mask advantage from geometry has not yet been demonstrated.

Change the teaching signal. Keep the comparison fair.

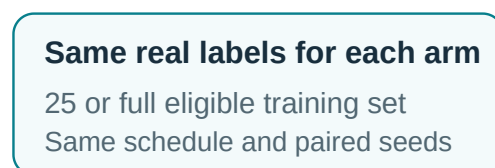
The extra geometric objective is applied only where the image-to-surface link is verified.

01 SYNTHETIC TRAINING

PROPOSED STUDY



02 REAL FINE-TUNING



03 DEPLOYMENT



The test: does 3D teaching improve real-photo masks at the same label budget?

Same images and capacity

Use the same encoder, mask/point heads, synthetic image list and real labels in matched conditions.

Only linked data teach 3D

Static synthetic images retain ordinary mask loss. No invented surface labels.

Real input stays simple

One photograph produces a visible defect mask. Physical depth and missing volume are separate extensions.

One image in. A visible mask and an inspectable output out.

This working viewer uses existing predictions to demonstrate the proposed interface.

MAIN JET / IMAGE REVIEW

AEBIS 329



● Saved prediction Reference layer off in this view.

OUTPUT RECORD · EXAMPLE SCHEMA

```
{
  "image_id": "AEBIS_329",
  "model": "existing_real_only",
  "input": "rgb_only",
  "predicted_pixels": 705,
  "mask_grid": [384, 288],
  "status": "saved_prediction_replay"
}
```

The deliverable: an RGB image, a predicted binary damage mask, an overlay and a model/output record. Reference masks are available for evaluation, not required by the deployed model.

The final geometry-trained model would use this same output format. Its real performance is currently unmeasured. No physical depth, severity or maintenance decision is inferred by this demo.

Five conditions answer five different questions.

The decisive comparison is G versus S and D at the same real-label budget.

ARM	TRAINING PATH	WHAT IT CHECKS
R	Real only Fixed pretrained backbone → real damage labels.	Strong baseline without synthetic task training.
R+	More real updates Same real subset, with extra optimization updates.	Whether additional optimization explains an apparent gain.
S	Synthetic + real Static and linked RGB with 2D labels → real labels.	Whether the synthetic stage itself adds value.
G	Geometry + real Exactly S's images + verified 3D point loss → same real labels.	Whether geometry adds value beyond the same images.
D	2D distance + real Exactly S's images + image-space distance loss → same real labels.	Whether ordinary 2D spatial teaching explains the gain.

25 / 397

planned real-label budgets

3 seeds

paired across all arms

30

fine-tuned models: $5 \times 2 \times 3$

9

shared synthetic checkpoints

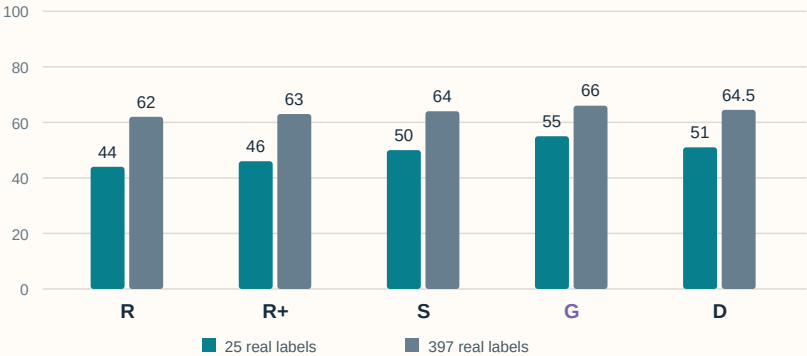
Reconfirm the full pool after the grouping audit. Use nested shared subsets, the same architecture/initialization, equal tuning allowance and matched image exposure. R+ matches update count; report measured runtime separately.

How the final comparison would read

An example of the reporting format—not an experimental result or performance forecast.

ALL VALUES ON THIS PAGE ARE INVENTED FOR THE DEMONSTRATION. The proposed study has no measured results yet.

ILLUSTRATIVE VALUES · NOT MEASURED



Invented mean IoU (%) at two real-label budgets. No uncertainty intervals are shown because no runs or independent samples produced these values.

ILLUSTRATION · ALL CELLS INVENTED

Arm	IoU 25 labels	IoU 397 labels	Healthy alarms 25 labels
R	44.0	62.0	4.0%
R+	46.0	63.0	4.2%
S	50.0	64.0	4.1%
G	55.0	66.0	4.7%
D	51.0	64.5	4.4%

IoU measures mask overlap. Alarm rate is the percentage of healthy images flagged under a fixed rule. These columns represent different evaluation populations.

Illustration: added geometry meets the numerical margins

At 25 labels, $G-S = +5.0$ pp and $G-D = +4.0$ pp. Healthy alarms rise by 0.6 pp versus S. But small-budget G is 7.0 IoU points below full-budget R, so this illustration does not meet the reduced-label margin. Real evidence and paired uncertainty would still be required.

Success means better real masks with controlled false alarms.

These are proposed planning margins, not achieved improvements.

ADDED GEOMETRIC VALUE

$$G - S \geq 3 \text{ pp}$$

At the small real-label budget

G must also beat D, the ordinary 2D distance control. Report paired uncertainty, not just one seed.

HEALTHY-IMAGE CONSTRAINT

$$\leq +2 \text{ pp}$$

Increase in healthy alarm rate

A larger predicted mask is not useful if it creates too many healthy-image warnings.

REDUCED-LABEL CLAIM

Within 2 pp

Of full-budget real-only IoU

Small-budget performance must approach full-budget R while satisfying the healthy-alarm constraint.

If the evidence supports geometry

Repeat on genuinely unused, grouped real images. Examine small defects, source differences and the number of annotation hours saved.

If geometry does not add value

Retain the strong real baseline. If S helps but G does not, claim a synthetic-data benefit only. Report negative or inconclusive outcomes.

Unmeasured today: real G-S, real G-D, geometry's healthy-alarm effect and label-efficiency benefit. Confidence intervals require actual paired results and enough independent evaluation groups.

SAY IT SIMPLY "A better number in simulation is encouraging. The decision depends on whether we find real damage more accurately and avoid false alarms."

The final report must show where the model fails.

Good masks, missed damage and healthy false alarms belong in the same evaluation.

ACTUAL SAVED ROBUSTNESS CHECK

Real-photo masks are sensitive to image quality



131 reused AEBIS development photographs · three retained models · fixed thresholds

Historical development evidence. Image corruptions do not replace evaluation on new physical blades.

Measured historical check: unchanged saved outputs replayed exactly; altered-image scores reveal sensitivity. This check evaluates the existing model; it does not improve it.

Group the evaluation correctly

Keep physical blades, capture sessions and video sequences separate when identities are available. Three seeds are three training repeats, not three blades.

Use truly unused real examples

The 131 AEBIS development images and exposed AeBAD tests are exploratory. Renaming a reused split does not make it fresh.

Add healthy examples

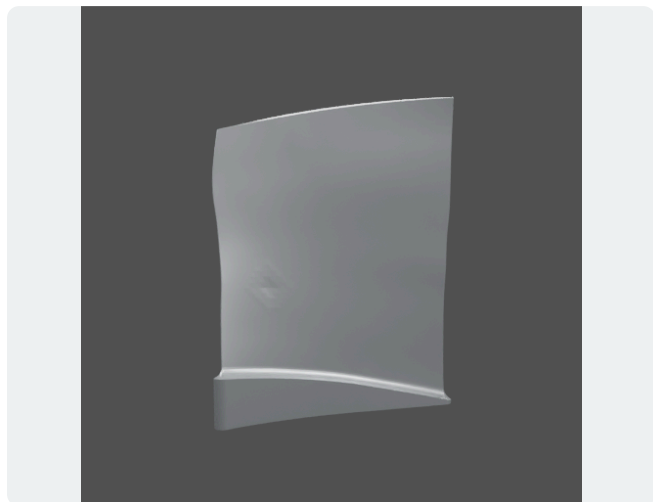
Measure false-marked area and healthy-image alarm rate. The damaged-only AEBIS pool cannot establish healthy specificity.

Final result package: paired comparisons + uncertainty, source/defect-size breakdowns, successes and misses, healthy controls, input/output records, and measured runtime.

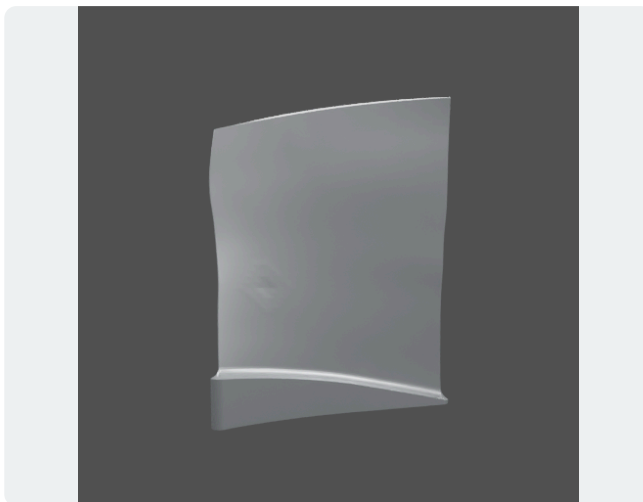
We also built physical-fitting tools. They answer a different question.

This saved example selects a simulated dent candidate using supplied geometry and acquisition settings.

(a) Observed generated image · obs_a



(b) Selected coarse candidate



(c) Selected fine candidate

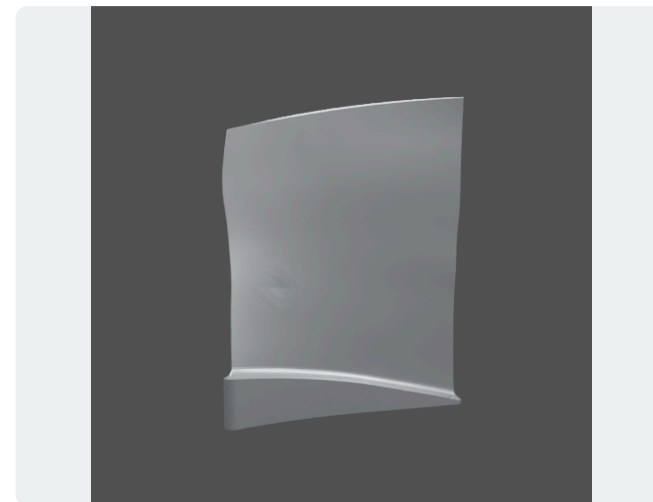


IMAGE SQUARED ERROR · LOWER IS BETTER

11,398 → 4,796

A closer image match within the prepared candidate family.

3D RMS ERROR · SAME 139-VERTEX POPULATION

$8.169 \times 10^{-5} \rightarrow 4.421 \times 10^{-5}$

Normalized source units. These are not millimetres.

What it establishes: a controlled fitting baseline with healthy CAD, candidate family and acquisition information supplied. This is nonlearned candidate selection on an exposed synthetic case. Reliable unknown physical recovery remains a separate extension.

A focused pilot, with a decision at the end.

The first milestone is a valid comparison protocol and eligible data—not another architecture.

WEEK 01 / DATA

Make the study valid

Audit duplicates, prior exposure, groups and mask meanings.

Freeze eligible synthetic lists, nested real subsets, endpoints and rules.

Gate A: data and protocol ready

WEEK 02 / SETUP

Verify and freeze

Replay baselines and verify the linked loss.

Run a technical runtime check, finish allowed tuning, freeze settings and build synthetic checkpoints.

Verified implementation + run budget

WEEK 03 / COMPARISON

Run the matched arms

Evaluate synthetic-only checkpoints. Fine-tune all five arms at both label budgets.

Check actual loaded image IDs, update counts and completion.

Gate B: matched and complete runs

WEEK 04 / DECISION

Evaluate and explain

Freeze predictions. Report real masks, uncertainty, healthy alarms and failures.

Package the prototype and decide: continue, revise or stop.

Evidence-backed research decision

What the team needs to resolve first

A consistent damage annotation guide, verified grouping and label eligibility, and access to genuinely unused damaged and healthy real images.

What will be delivered

A versioned data manifest, matched comparison tables and figures, a failure analysis, and an RGB-to-mask demonstration with saved outputs.

Four weeks is a planning estimate for one researcher, subject to annotation availability, fresh evaluation data and measured runtime. The nominal cap is nine synthetic-stage checkpoints plus 30 fine-tuned models.

Can extra knowledge during training reduce the real labelling burden?

That is the central question this proposal turns into a controlled experiment.

- 1 **We already have** synthetic and real data pipelines, retained image models, linked-label tools and independently checked examples.
- 2 **We have learned** that strong synthetic results and good 3D point placement do not automatically give good real masks.
- 3 **We will compare** the same images and model with different teaching signals, using identical real-label access.
- 4 **We will claim an improvement** only if it appears in the real task and survives the matched controls and healthy-image checks.

ONE-MINUTE SUPERVISOR SUMMARY

“We want to use known 3D damage in simulation to teach a model to find defects in real photographs.

We have working models and examples, but we have not established that geometry improves real segmentation. The next study isolates that question with five matched conditions.”

The proposed contribution: evidence about when geometric supervision helps synthetic-to-real damage localization, together with a reproducible method, fair controls and a usable RGB output.

Traceable examples. Clear limits.

This presentation was built from the uploaded proposal and saved local evidence on 22 September 2026.

[P] Uploaded research proposal · v1.1

Primary task scope, proposed method, five arms, label budgets, endpoints and four-week plan.

`MAIN_JET_RESEARCH_PROPOSAL.pdf`

[E1] Retained real model and saved outputs

397 training / 131 reused development images. Mean IoU recomputed from saved per-image scores to 1.11×10^{-16} agreement. Cases 329 and 486 use seed 20260915.

`experiments/blade_diagnostics/blade_missed_sampling_v1/20260915T113013Z-64a8a0b61bd4/`

Exact current source paths are recorded in the accompanying evidence manifest.

[E3] Existing-model robustness check

Same retained models, fixed corruptions, 131 reused AEBIS development images. Zero new training for that check.

`reports/BLADE_REAL_OVERNIGHT_RUN.md`

`experiments/blade_diagnostics/blade_real_robustness_v1/20260917T152254Z-7b61b9c9e86a/RESULTS.md`

[E5] Controlled physical-fitting example

Fine/coarse obs_a comparison on the same 139-vertex population, with supplied healthy CAD, candidates and acquisition settings.

`reports/BLADE_FINE_BANK_RESULTS_20260922.md`

`experiments/blade_diagnostics/blade_fine_bank_fit_v1/20260922T091750Z-65708eedaf72/`

[E0] Project evidence audit · 22 September

Inventory, exposure history, examples and source receipts. This demo uses only relevant proposal evidence.

`reports/full_status_audit_20260922T101357Z/audit/{inventory_counts,results_examples,results_families}.json`

[E2] Matched synthetic 2D / 3D teaching

Same RGB models; three paired seeds; known geometry for training/evaluation. Nick p5 display crops are chosen using the reference, only for readability.

`reports/BLADE_VARIED_3D_RESULTS.md`

`experiments/blade_diagnostics/blade_varied_3d_v1/20260917T082235Z-8aafa30da222/`

[E4] Broad synthetic-to-real transfer

Saved broad synthetic and real triptychs; dataset-wide scores belong to different populations.

`reports/BLADE_BROAD_DATA_RESULTS.md`

`experiments/blade_diagnostics/blade_broad_data_evaluation_v1/20260914T140257Z-ef7392ed729f/gallery.json`

How to read and share this demo

Teal = reference; amber = prediction; purple = geometric supervision. Labelled mock results are invented solely to demonstrate reporting. No training or new inference was performed.

HTML works offline with embedded assets. PDF preserves the presentation's default examples and illustrative scenario. Arrow keys navigate; Escape exits presentation mode.